

THE RELATIONSHIP BETWEEN MOBILE PHONE USAGE AND SLEEP QUALITY AMONG STUDENTS

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COURSE: MATHEMATICAL STATISTICS WITH COMPUTING| STQD6214



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BACKGROUND

This case study explores the implications of mobile phone overuse on students' well-being, highlighting the relationship between bedtime phone habits and disrupted sleep patterns.

Research Question:

How does daily mobile phone usage affect students' sleep quality?

Objectives:

- Describe students' mobile phone usage patterns
- Assess students' sleep quality
- Examine the relationship between phone usage and sleep quality
- Evaluate result reliability using Monte Carlo simulation

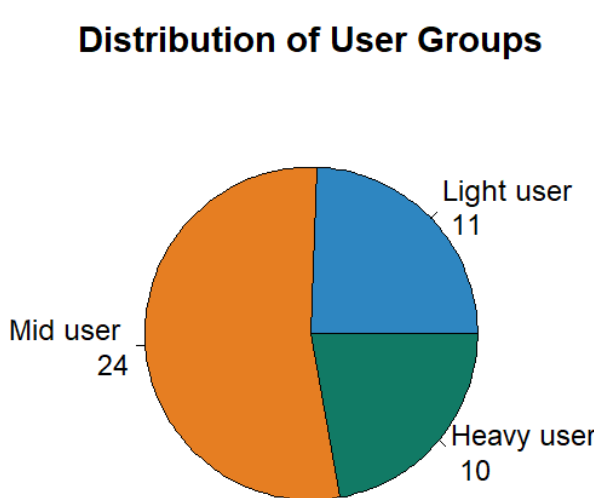
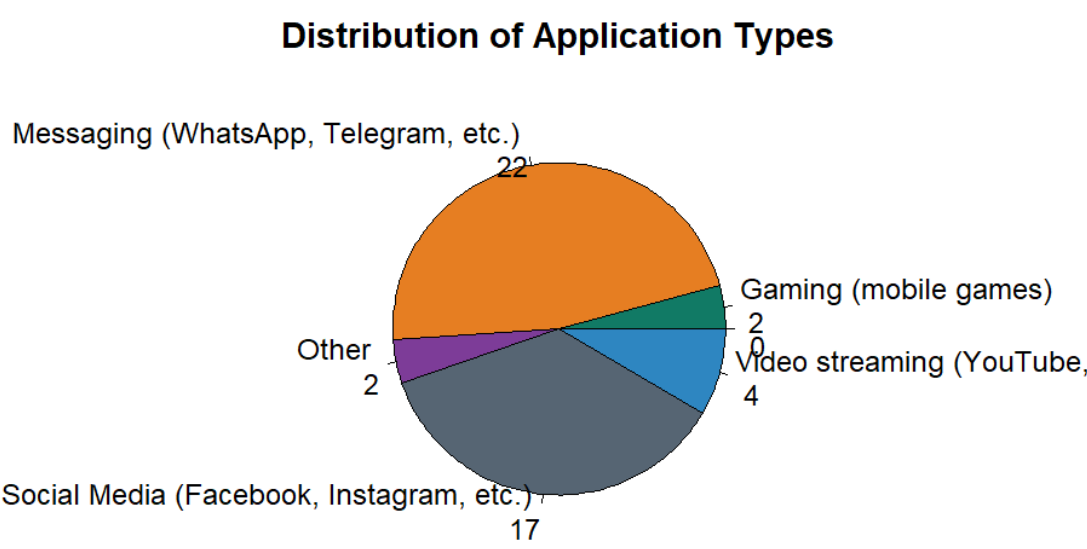
Importance of the Study:

Poor sleep affects academic performance, mental health, and well-being. Understanding whether phone use contributes to poor sleep can help students make healthier lifestyle choices.

Data Collection:

was designed in XLS form survey (Malay and English), uploaded to the KOBO server, and distributed to students at UKM University. Survey link:

<https://ee.kobotoolbox.org/x/7v2OW2mb>



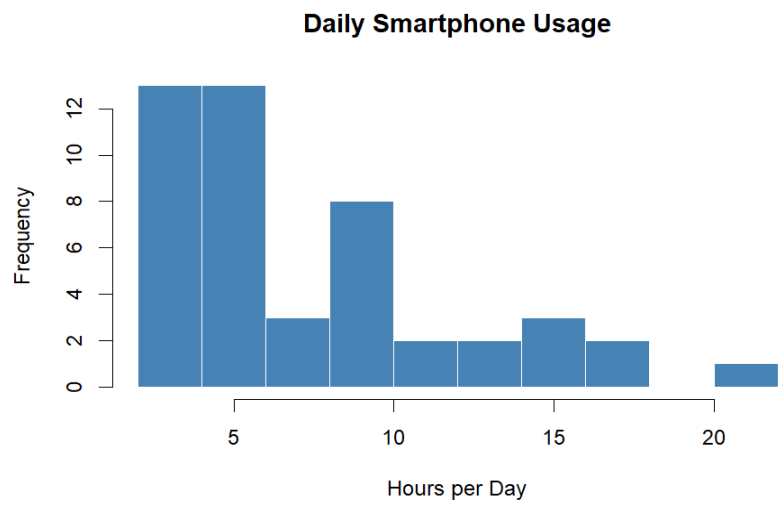
EXPLORATORY DATA ANALYSIS

Data Preparation:

As a first step, we performed data cleaning and transformation using R (selecting relevant columns, renaming variables, converting variables type, handling missing values).

Descriptive Statistics:

Daily Smartphone Usage :
Mean = 7.8, SD = 4.8
Sleep Quality Score:
Mean = 2.04, SD = 0.72

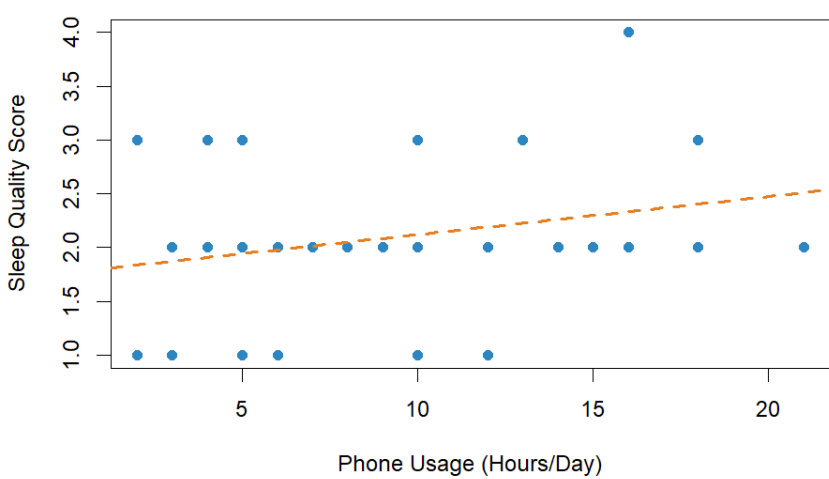


Insight:

Students use smartphones an average of 7.8 hours/day (median 6), with most between 4–10 hours and a few extreme heavy users. Sleep quality averages 2.0 (fairly good), with most clustered at score 2 and very poor sleep uncommon.

HYPOTHESIS TESTING & REGRESSION

Linear Regression Model:
 $Sleep\ Quality = 1.77 + 0.035 \times Phone\ Hours$
Interpretation
Higher phone use slightly increases poor sleep score
Effect is weak and not statistically significant



Hypothesis Test 2: Sleep Quality
 H_0 : Mean sleep quality score = 2 (normal)
 H_1 : Mean $\neq$ 2
Result: p-value = 0.6876 $\rightarrow$ Fail to reject H_0
 $\rightarrow$ Sleep quality is statistically close to "fairly good".

Hypothesis Test 1: Phone Usage
 H_0 : Average phone usage = 6 hours/day
 H_1 : Average phone usage > 6 hours/day
Result: p-value = 0.0066 $\rightarrow$ Fail to reject H_0
 $\rightarrow$ No strong evidence that average usage is significantly above 6 hours.

Correlation Analysis
Pearson correlation $r \approx 0.23$
Weak positive relationship

Monte Carlo Simulation

Monte Carlo simulation used in this to examine the stability of statistics by resampling the data 1,000 times.

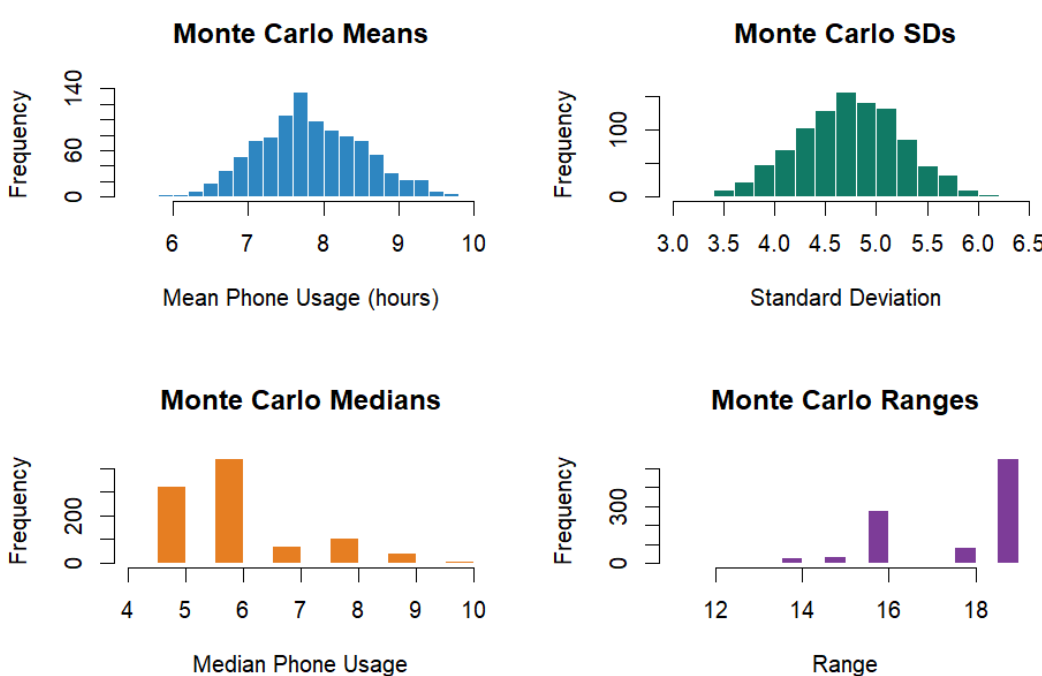


Parameters Simulated

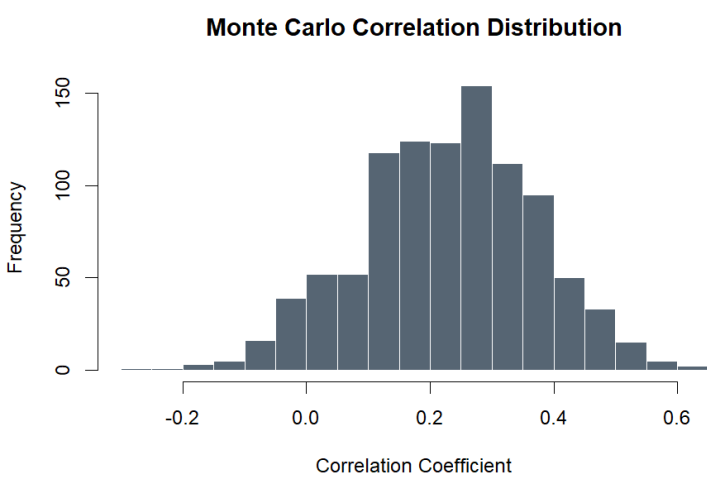
- Mean phone usage
- Standard deviation
- Median
- Range
- Correlation coefficient

FINDING

- Sampling distributions are approximately normal
- Mean phone usage remains stable around 7–8 hours
- Correlation values cluster near 0.2
- Confirms that the observed weak relationship is consistent



The Monte Carlo simulation (1,000 resamples) shows that the correlation between phone usage and sleep quality is centered close to zero, with values spread weakly in both directions.



CONCLUSION

While heavy smartphone use is common, its impact on sleep quality among students appears weak in this sample.